

Bootstrap Tables — Complete Notes

Introduction to Bootstrap Tables

What is a Table?

An HTML table is used to display data in rows and columns.

Tables are useful when information needs to be organized and compared.

Common examples include:

- Student records
- Employee records
- Product lists
- Exam results
- Attendance
- Price lists
- Sales reports
- Order details
- Customer information
- Financial data

Example of a simple HTML table:

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>Course</th>
  </tr>

  <tr>
    <td>Rahul</td>
    <td>20</td>
    <td>HTML</td>
  </tr>

  <tr>
    <td>Priya</td>
    <td>21</td>
    <td>CSS</td>
  </tr>
</table>
```

The basic HTML table elements are:

Element	Purpose
<table>	Creates the table
<tr>	Creates a table row
<th>	Creates a table heading cell
<td>	Creates a table data cell
<thead>	Contains the table header
<tbody>	Contains the main table data
<tfoot>	Contains the table footer

What are Bootstrap Tables?

Bootstrap provides predefined CSS classes that can be applied to normal HTML tables to make them look cleaner and more professional.

Instead of writing CSS manually, we can use Bootstrap's table classes.

The most important class is:

`.table`

Example:

```
<table class="table">
  <thead>
    <tr>
      <th>Name</th>
      <th>Age</th>
      <th>Course</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Rahul</td>
      <td>20</td>
      <td>HTML</td>
    </tr>

    <tr>
      <td>Priya</td>
      <td>21</td>
      <td>CSS</td>
    </tr>
  </tbody>
</table>
```

The `.table` class provides Bootstrap's basic table styling.

Setting Up Bootstrap

Before using Bootstrap table classes, Bootstrap CSS must be included in the HTML document.

Using Bootstrap CDN

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>Bootstrap Tables</title>

  <link
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
    rel="stylesheet">

</head>

<body>

</body>

</html>
```

Important

The viewport meta tag is important for responsive behavior:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Basic Bootstrap Table

The simplest Bootstrap table uses:

```
.table
```

Example:

```
<table class="table">

  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Course</th>
  </tr>

  <tr>
    <td>1</td>
    <td>Rahul</td>
    <td>HTML</td>
  </tr>

  <tr>
    <td>2</td>
    <td>Priya</td>
    <td>CSS</td>
  </tr>

  <tr>
    <td>3</td>
    <td>Amit</td>
    <td>Bootstrap</td>
  </tr>

</table>
```

What does `.table` do?

The `.table` class provides Bootstrap's basic table styling.

It improves:

- Spacing
- Borders
- Alignment
- Typography
- Overall readability

Table Borders

Bootstrap provides border utilities for tables.

The most common class is:

`.table-bordered`

Example:

```
<table class="table table-bordered">
```

```
  <thead>
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
    </tr>
  </thead>
```

```
<tbody>
```

```
  <tr>
    <td>1</td>
    <td>Rahul</td>
    <td>HTML</td>
  </tr>
```

```
  <tr>
    <td>2</td>
    <td>Priya</td>
    <td>CSS</td>
  </tr>
```

```
</tbody>
```

```
</table>
```

Borderless Tables

Bootstrap also provides:

```
.table-borderless
```

Example:

```
<table class="table table-borderless">
```

```
  <thead>
    <tr>
      <th>Name</th>
      <th>Age</th>
    </tr>
  </thead>
```

```
<tbody>
```

```
  <tr>
    <td>Rahul</td>
    <td>20</td>
  </tr>
```

```
  <tr>
    <td>Priya</td>
    <td>21</td>
  </tr>
```

```
</tbody>
```

```
</table>
```

This removes the table's borders.

When should we use it?

Borderless tables can be useful for:

- Simple information lists
- Minimal interfaces
- Dashboard sections
- Clean designs

Table Hover Effect

Bootstrap provides:

```
.table-hover
```

This creates a hover effect when the user moves the mouse over table rows.

Example:

```
<table class="table table-hover">

  <thead>
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
    </tr>
  </thead>

  <tbody>

    <tr>
      <td>1</td>
      <td>Rahul</td>
      <td>HTML</td>
    </tr>

    <tr>
      <td>2</td>
      <td>Priya</td>
      <td>CSS</td>
    </tr>

    <tr>
      <td>3</td>
      <td>Amit</td>
      <td>Bootstrap</td>
    </tr>

  </tbody>

</table>
```

When the mouse moves over a row, Bootstrap applies a hover style.

Striped Rows

Bootstrap provides:

```
.table-striped
```

This applies alternating background styling to table rows.

Example:

```
<table class="table table-striped">

  <thead>
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Marks</th>
    </tr>
  </thead>

  <tbody>

    <tr>
      <td>1</td>
      <td>Rahul</td>
      <td>85</td>
    </tr>

    <tr>
      <td>2</td>
      <td>Priya</td>
      <td>92</td>
    </tr>

    <tr>
      <td>3</td>
      <td>Amit</td>
      <td>78</td>
    </tr>

  </tbody>

</table>
```

Conceptually:

```
+-----+
| ID | Name | Marks |
+-----+
| 1  | Rahul | 85    |
+-----+
| 2  | Priya | 92    |
+-----+
| 3  | Amit  | 78    |
+-----+
```

The alternating row styling makes large tables easier to read.

How `.table-responsive` Works

The responsive wrapper provides horizontal scrolling when the table is wider than the available container.

Conceptually:

Desktop:

```
+-----+
| ID | Name | Email | Phone | Address | Course |
+-----+
```

Mobile:

```
+-----+
| ID | Name | Email | -->|
+-----+
      |
      | Horizontal scrolling
      v
    More columns
```

Important

`.table-responsive` is usually applied to a wrapper around the `<table>`, not directly to the `<table>`.

Correct:

```
<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

Responsive Table at Specific Breakpoints

Bootstrap also provides breakpoint-specific responsive table wrappers.

Examples:

```
.table-responsive-sm
.table-responsive-md
.table-responsive-lg
.table-responsive-xl
.table-responsive-xxl
```

Example:

```
<div class="table-responsive-md">

  <table class="table">

    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Email</th>
        <th>Phone</th>
      </tr>
    </thead>

    <tbody>
      <tr>
        <td>1</td>
        <td>Rahul</td>
        <td>rahul@example.com</td>
        <td>9876543210</td>
      </tr>
    </tbody>

  </table>

</div>
```

The responsive wrapper becomes active below the specified breakpoint.

Responsive Table Classes

Class	Purpose
.table-responsive	Responsive behavior at all viewport sizes
.table-responsive-sm	Responsive below sm
.table-responsive-md	Responsive below md
.table-responsive-lg	Responsive below lg
.table-responsive-xl	Responsive below xl
.table-responsive-xxl	Responsive below xxl

Bootstrap Table Class Cheat Sheet

Class	Purpose
<code>.table</code>	Basic Bootstrap table
<code>.table-striped</code>	Alternating row styling
<code>.table-hover</code>	Hover effect on rows
<code>.table-bordered</code>	Adds borders
<code>.table-borderless</code>	Removes borders
<code>.table-sm</code>	Compact table
<code>.table-dark</code>	Dark table
<code>.table-primary</code>	Primary contextual style
<code>.table-secondary</code>	Secondary contextual style
<code>.table-success</code>	Success contextual style
<code>.table-danger</code>	Danger contextual style
<code>.table-warning</code>	Warning contextual style
<code>.table-info</code>	Info contextual style
<code>.table-light</code>	Light contextual style
<code>.table-responsive</code>	Responsive table wrapper
<code>.table-responsive-sm</code>	Responsive wrapper below <code>sm</code>
<code>.table-responsive-md</code>	Responsive wrapper below <code>md</code>
<code>.table-responsive-lg</code>	Responsive wrapper below <code>lg</code>
<code>.table-responsive-xl</code>	Responsive wrapper below <code>xl</code>
<code>.table-responsive-xxl</code>	Responsive wrapper below <code>xxl</code>
<code>.caption-top</code>	Places caption above the table
<code>.text-start</code>	Start-aligns text
<code>.text-center</code>	Centers text
<code>.text-end</code>	End-aligns text
<code>.align-top</code>	Top-aligns content
<code>.align-middle</code>	Vertically centers content
<code>.align-bottom</code>	Bottom-aligns content

Questions and Answers

Q1. What is a Bootstrap table?

A Bootstrap table is a normal HTML table styled using Bootstrap's predefined table classes.

Q2. Which class is used to create a basic Bootstrap table?

`.table`

Q3. What does `.table-striped` do?

It applies alternating styling to table rows to improve readability.

Q4. What does `.table-bordered` do?

It adds borders around the table cells.

Q5. What does `.table-hover` do?

It applies a hover effect when the user moves the pointer over table rows.

Q6. What does `.table-sm` do?

It creates a more compact table by reducing cell padding.

Q7. What is `.table-dark`?

It applies a dark theme to the table.

Q8. What is `.table-borderless`?

It removes borders from the table.

Q9. What is `.table-responsive`?

It provides a horizontally scrollable table wrapper when the table is wider than the available space.

Q10. Where should `.table-responsive` normally be applied?

It is normally applied to a wrapper around the `<table>`.

Example:

```
<div class="table-responsive">

  <table class="table">
    ...
  </table>

</div>
```

Q11. Can Bootstrap table classes be combined?

Yes.

Example:

```
<table class="table table-striped table-hover table-bordered">
```

Q12. What is the difference between `.table` and `.table-striped`?

`.table` provides the basic Bootstrap table styling.

`.table-striped` adds alternating row styling.

Q13. What is the difference between `.table-bordered` and `.table-borderless`?

`.table-bordered` adds borders.

`.table-borderless` removes borders.

Q14. Can contextual table classes be applied to rows?

Yes.

Example:

```
<tr class="table-success">
```

Q15. Can contextual classes be applied to individual cells?

Yes.

Example:

```
<td class="table-danger">  
    Failed  
</td>
```

Q16. What is the purpose of `<thead>` ?

It contains the header rows of a table.

Q17. What is the purpose of `<tbody>` ?

It contains the main data rows.

Q18. What is the purpose of `<tfoot>` ?

It contains summary or footer information.

Q19. Why should `<th>` be used instead of `<td>` for headings?

`<th>` semantically identifies a table header and provides better structure and accessibility.

Q20. Should tables be used for webpage layout?

No.

Tables should primarily be used for tabular data. Bootstrap's grid system should be used for page layout.

47. Quick Revision

Remember the most important Bootstrap table classes:

```
.table
.table-striped
.table-hover
.table-bordered
.table-borderless
.table-sm
.table-dark
.table-primary
.table-secondary
.table-success
.table-danger
.table-warning
.table-info
.table-light
.table-responsive
.caption-top
```

Basic Table

```
<table class="table">
```

Striped Table

```
<table class="table table-striped">
```

Hover Table

```
<table class="table table-hover">
```

Bordered Table

```
<table class="table table-bordered">
```

Borderless Table

```
<table class="table table-borderless">
```

Compact Table

```
<table class="table table-sm">
```

Dark Table

```
<table class="table table-dark">
```

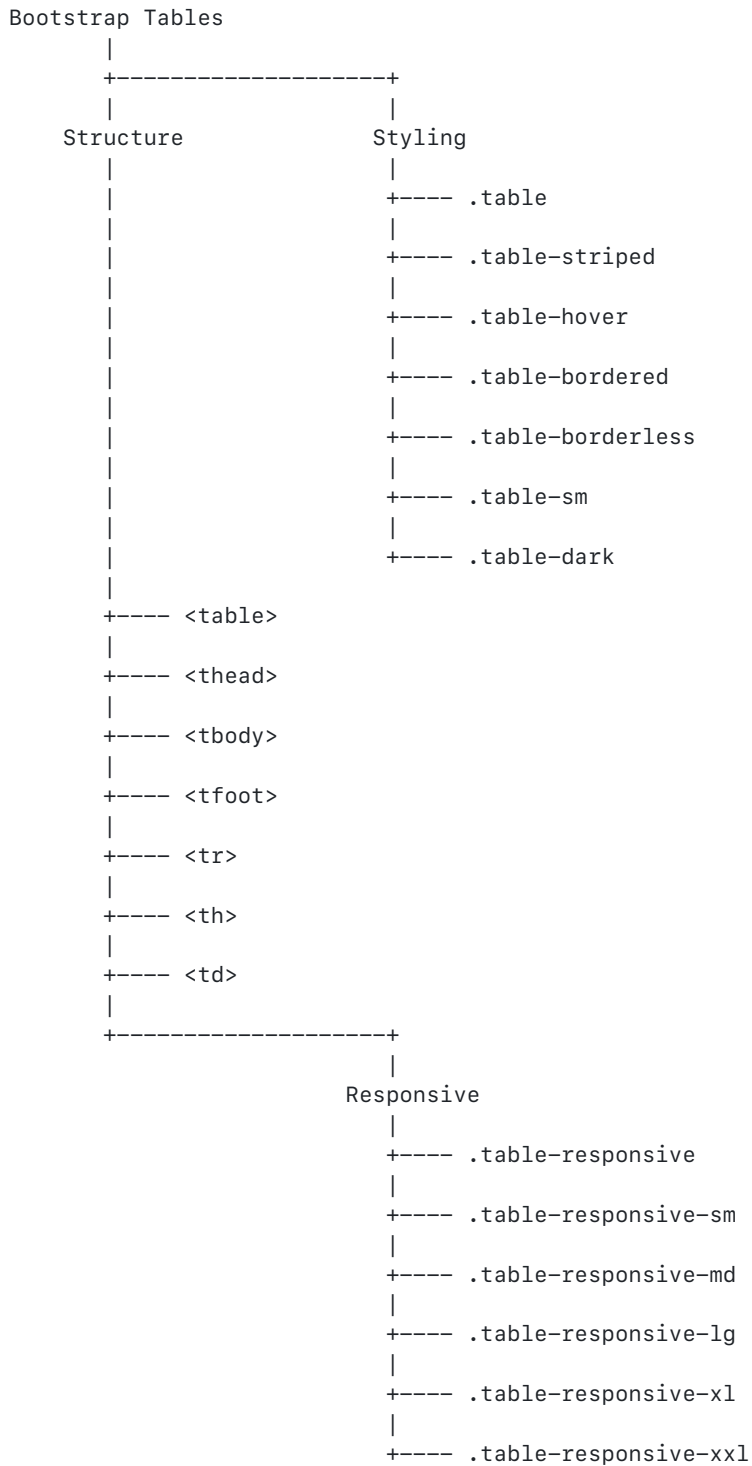
Responsive Table

```
<div class="table-responsive">  
  <table class="table">  
    ...  
  </table>  
</div>
```

Multiple Styles

```
<table class="table table-striped table-hover table-bordered">
```

48. Final Concept Map



49. Final Summary

Bootstrap Tables make it easier to create clean, readable, and responsive tables using predefined Bootstrap classes.

The most important concepts to remember are:

1. Bootstrap tables are still normal HTML tables.
2. Use `.table` as the basic Bootstrap table class.
3. Use `.table-striped` for alternating row styling.

4. Use `.table-hover` for row hover effects.
5. Use `.table-bordered` for borders.
6. Use `.table-borderless` to remove borders.
7. Use `.table-sm` for compact tables.
8. Use `.table-dark` for a dark table.
9. Use contextual classes such as `.table-success` , `.table-danger` , and `.table-warning` to communicate states.
10. Use `.table-responsive` when a table may be wider than the screen.
11. Use `<thead>` , `<tbody>` , and `<tfoot>` to organize table content.
12. Use `<th>` for headers.
13. Use `scope` attributes where appropriate for accessibility.
14. Use tables for tabular data, not for creating the overall webpage layout.
15. Use Bootstrap's Grid System for webpage layout.
16. Bootstrap table classes can be combined to create different table designs.

Most Important Pattern to Remember

```
<div class="table-responsive">

  <table class="table table-striped table-hover table-bordered">

    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">Name</th>
        <th scope="col">Course</th>
      </tr>
    </thead>

    <tbody>

      <tr>
        <th scope="row">1</th>
        <td>Rahul</td>
        <td>HTML</td>
      </tr>

      <tr>
        <th scope="row">2</th>
        <td>Priya</td>
        <td>CSS</td>
      </tr>

    </tbody>

  </table>

</div>
```

This pattern combines the fundamental concepts required to create a structured, styled, accessible, and responsive Bootstrap table.